Alexandria Engineering Journal 127 (2025) 817–829

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

Original article

Design of novel intelligent electronic trap for early detection and monitoring of tomato crops pest *Tuta Absoluta* using Deep learning

Yaser M. Abid Alasady [a,b], Eduardo Pérez [a], Sebastián Ventura [a,*]

[a] Andalusian Research Institute in Data Science and Computational Intelligence (DaSCI). University of Córdoba, Córdoba, Spain
[b] Business Informatics (BIC). University of Information Technology and Communications (UoITC), Baghdad, Iraq

ARTICLE INFO

Dataset link: alasady-tuta-absoluta-dataset

Keywords:
Electronics system
Deep learning
Pest detection
Intelligent system
Crop control

ABSTRACT

To control insect pests and reduce the destruction of agricultural crops, the process of detection and monitoring of pests is an urgent need at present time. Due to the tremendous development in technology, the traditional methods used in laboratories considered a waste of time and human efforts. In this research, a new data set collected and published for the first time, novel electronic trap designed with intelligent system to detect pests in tomato crop, and monitor the spread of the pest periodically and continuously based on the collected dataset. As the designed intelligent system firstly consists of a novel trap designed in a way that contains six colored sticky traps to catch insects continuously, controlled by L293D IC to rotate the motor, a digital camera used to provide the system with real images at periodic intervals. To detect pest, the (YOLOv11, YOLOv9, YOLOv8 and YOLOv5) used for this purpose. The Tuta Absoluta pest used for the detection and monitoring process of the designed novel intelligent system. The results used in the system; the precision was 95.9%, recall was 92.5%, Mean Average Precision (mAP 0.5) was 94% and F1 score was 94% and the results were promising. As compared to other models of (YOLOv11, YOLOv9, YOLOv8 and YOLOv5), the YOLOv5x shows that its higher results than other models. This system is easy to use and accurate in providing the information required monitoring the spread of the insect pest, therefore it could use in modern agricultural applications.

1. Introduction

One of the most basic requirements for human life is the provision of clean and sufficient food. The number of the world's population is expected to keep growing and increase by 1.1% every year [1], and these statistics increase the need to provide healthy food that covers human needs. In many countries of Africa, more than 64% of crops were consumed in 2016 by insect pests [2] and nearly half of the global product is destroyed annually by diseases affecting crops or by various insect pests [3]. One of the easy ways to control insect pests that are widely used in most countries of the world was pesticides. since most farmers do not have sufficient and necessary information and the correct method of control [1,4]. Sometimes small quantities of pesticides are sprayed and that are not enough to eliminate all insects in the field and it is very contained. For these insects that were not killed by pesticides to produce an entire generation resistant to the pesticides to which they were previously exposed. in other cases; the plants are sprayed with pesticides in very large quantities to ensure the complete and actual elimination of pests completely, but in return it will be accompanied by great material losses in addition to the possibility of human life exposure to various diseases, especially cancerous diseases [5]. IPM has been proposed to contribute to reducing costs and preserving the environment as much as possible through continuous monitoring of insect pests [6–8]. It is very clear that these traditional methods contain many disadvantages. The most important disadvantage was the continuous monitoring, which is consuming time and effort, in addition to the completely manual diagnosis, which can be incorrectly diagnosed, in addition to the laboratory costs to detect pests and infections in plants [4,9–11]. Because of these problems, there is an urgent need for a method that can facilitate the process of controlling the spread of diseases and pests automatically. Fortunately, technology is in tremendous development, especially in areas where artificial intelligence plays a key role, which can be used in applications and follow-up of agricultural crops [12–15].

In this research, a novel trap will designed based on artificial intelligence for early monitoring of the spread of pests in order to control the pest before it reaches the stage of danger, which becomes difficult to control, and in this research several tasks implemented:

* Corresponding author.
E-mail address: sventura@uco.es (S. Ventura).
URL: https://www.uco.es/users/sventura/ (S. Ventura).

https://doi.org/10.1016/j.aej.2025.06.054
Received 27 February 2025; Received in revised form 2 May 2025; Accepted 20 June 2025
Available online 8 July 2025
1110-0168/© 2025 The Authors. Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Novel dataset was collected and published through this research after conducting many experiments and can be applies to many different ideas.
2. Proposed the design an intelligent electronics trap that helps reduce the human effort spent on detecting pests with very high accuracy.
3. The intelligent electronics trap provide the required information on a daily, weekly, or monthly information as needed.
4. Different Artificial intelligence models for pest detections in the trap used to accurately, detect the type of pest and the number of insects expected in the field.
5. Early detection of pest helps directly to make the right decisions to limit or reduce the spread of the pest.

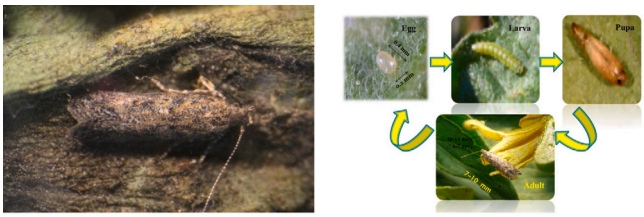
In this research, we will address and organized as follows: a study of related works in Section 2, research methodology, which contains the description of Dataset and the design of the intelligent electronic trap and how to add and trained artificial intelligence to detect pests in an early manner described in Section 3; Section 4 described the annotation and preparation of dataset collected for this research; Section 5 represent a detailed of experimental study and the results obtained; the discussion represented in Section 6 and the conclusion of this research represented in Section 7.

2. Related works

Artificial intelligence still considered a great challenge to enter into all aspects of life, being an essential part of human technological life, and artificial intelligence plays a very important role in the development of agriculture during the previous years. [16] They identifying soybean leaf and herbivorous pest by using Convolution Neural Network using different training strategies with the help of semi-supervised methodologies to classify an unlabeled training set used in the training. [17] Used (Resnet-50, Xception, VGG-16, VGG-19 and Inception-v3) for classification of soybean pest images in order to makes pest control management. [18] Applying CNN to recognize ten species of crop pests and GoogLeNet model fine-tuned used to deal with unwanted backgrounds in images. [19] Separated grains by traditional methods takes a lot of time and effort, so this research suggested segmentation of grain types and divided them into three segments (complex-touching grains, non-touching grains and simple touching grains) by using K-means clustering algorithm, layered watershed algorithm and splitting line detection algorithm. [20] Works on calculating the density of cotton crops with the helping of CenterNet, using the faster-RCNN was able to detect cotton crop with a distance of half a meter between the crops and the camera. [21] Weeds are a very important reason for the loss of many agricultural crops, so use unmanned aerial vehicles to take pictures with high efficiency and using of Developed new weed segmentation model with residual neural networks. [1] Based on Support Vector Machines (SVMs), Paraconsistent Feature Engineering (PFE), Bark Scale (BS) and Wavelet-packet Transform (WPT), the acoustic detection of cicadas in plantations implemented. [22] Multiple species of pest detected based on cascading pest detection approach (DeepPest), they proposed two-stage mobile vision In order to be more accurate in detecting objects [4]. Proposed anchor-free region convolutional neural network (AF-RCNN) for recognition and classification of 24 types of pests. [23] collected images for 24 pests destroy field crops in China every year and they used YOLOv3, Faster RCNN, Cascade R-CNN and SSD to detect the pests in order to monitoring field crop pests. [24] Collect datasets for ten species of butterfly and for the identification, the Convolutional Neural Networks (CNNs), used between ten species. [11] The identification of ten species of crop pests with complex background achieved by using deep residual learning and then compared the result with back propagation neural networks and support vector machine. [25] Proposed system for detection of grapevine yellows symptoms in red grape vine by using six deferent methods (ResNet-101, Inception v3, AlexNet, GoogLeNet, ResNet-50, and SqueezeNet) and find that ResNet-50 have best results of accuracy in detection. [26] Make the detection more accurate proposed super resolution model based on deep recursive residual network for image reconstruction. [27] Classify lettuce leaf into four categories affected by light stress using hierarchical fusion convolutional neural network architecture (MFC-CNN).

Tomato was one of the most important and most cultivated vegetable crops in the world [28], in 2010 its production was estimated at 145.6 million tons, and there is still a need for cultivation in larger quantities. Spain, Italy, and Brazil countries are among the most important countries that produce tomato crops [29]. Tomato crops exposed to many risks because of climatic changes throughout the year, so many researchers went to add artificial intelligence to monitor diseases and risks that affect tomato crops. [30] Employed artificial intelligence using VGG16 and SVM to detect the presence of ten types of diseases that affect tomatoes. Which are Tomato bacterial spot, Tomato early blight, Tomato late blight, Tomato leaf mold, Tomato septoria leaf spot, Tomato two spotted spider mite, Tomato target spot, Tomato mosaic virus, Yellow leaf curl virus and Tomato gray spot. Depending on the condition of the tomato leaves, whether they are infected or healthy. [31] R-CNN used to locate insects in a tomato by misleading diseased areas. [32,33] for detection and identification of tomato gray leaf spot YOLO V3 was used and tomato leaves were monitored to detect the exposure of crops to the disease. [34] Used a large set of images with the help of CNN to reveal nine diseases of tomato leaves (Tomato yellow leaf curl virus, Tomato mosaic virus, Target spot, Spider mites, Septoria spot, Leaf mold, Late blight, Early blight and Bacterial spot). [35] CNN was trained on twenty-five types of diseases that affect plants, including tomato, where deep learning was provided with a set of pictures of infected and healthy tomato leaves for (Bacterial spot, Early blight, Late blight, Septoria leaf spot, Spider mites, Tomato mosaic virus, Leaf mold and Target spot) diseases. [36] Used CNN and many methods (AlexNet, GoogleNet and ResNet) to classify diseases that affect tomato leaves and after conducting the experiments, ResNet was the best in giving accurate information. In [37] research, YOLOv5 used to help collect tomato crops in order to reduce the effort expended in the process of collecting tomato crops by training YOLOv5 on the types of tomatoes that should be selected and suitable for collection among the collected crops. In this research [38], the improved YOLOv7-based multi-task deep convolutional neural network (DCNN) detection model used in order to help reduce the costs of using workers to harvest ripe tomatoes. [39] focus on reducing the damage that can happens to tomato fruits or tomato plant stems, this is because the robot responsible for collecting crops makes mistakes in determining the location of the tomatoes to be collected because tomato plants grow randomly and irregularly, For this reason, YOLOv8 used to detect and segment bodies and sepals of tomatoes. In [40] the FATDNet artificial intelligence model was proposed and used to detect diseases on tomato leaves. One of the most important obstacles they faced was the presence of a complex background or the small size of the diseases.

One of the deadly pests that consume approximately 80%–100% of tomato crops is the *Tuta Absoluta* pest as shown in Fig. 1(a). *Tuta Absoluta* pest goes through four stages during its life cycle, as shown in Fig. 1(b). The adult lays eggs on tomato leaves and the eggs hatch to turn into the second phase of her life period to become larva, at this stage of her life period it makes tunnels inside the leaves to provide them with energy and food, then the larva turns to the pupa, which in turn will turn into the last stage of the development of the insect, which is adults, and these age stages need different times depending on the temperature to transform from the egg stage to the adult stage, as in Table 1. The adult stage in the life cycle of the insect is very dangerous to crops, as the adult female can lay 250–300 eggs, which will pose a real threat in crop damage estimated at 80%–100% [41, 42]. Fig. 2. shows tomato crops affected by *Tuta Absoluta* pest. Many researchers have classify the infections of tomato crops using artificial



(a) *Tuta Absoluta* (b) *Tuta Absoluta* biology

Fig. 1. *Tuta Absoluta* Pest.

Table 1
Tuta Absoluta life cycle.

T	Egg	Larva	Pupa	Adult	Total life (day)
30°	4	11	5	9	29
15°	10	36	20	23	89

intelligence. [43] examined the infestations in the leaves to detect Tuta Absoluta pest, the infestation level was evaluated into three levels (No Tuta, Low Tuta and High Tuta) CNN approach and four methods used to classify the presence of a pest infestation (VGG16, VGG19, ResNet and Inception-v3) and after conducting the training it was found that Inception-v3 was the best in accuracy give information. [2] For early detection of an infestation in the leaves of Tuta Absoluta, they used three methods (VGG16, VGG19 and ResNet50) and after making comparisons, it found that VGG16 is the best in terms of accuracy in diagnosing the infection in the leaves. [44] In this study, images of tomato crops collected showing leaves infected with Tuta absoluta and Leveillula taurica and proposed the use of FASTER-RCNN to identify the location of leaf infestations, enabling early detection to minimize the spread of the pest. In [45], it was proposed to use YOLOv8l-Seg to detect the presence of Tuta absoluta pest in tomato leaves, as it is a dangerous and rapidly spreading pest, and measures must be taken quickly to reduce damage to tomato crops. It is possible to distinguish between images of healthy tomato leaves and tomato leaves infected with the pest. Deep learning commonly used in crop image detection due to its fast and convenient advantages, but YOLOV5 approaches are still extremely challenging in dealing with this kind of detection pest. Therefore, in this research, *Tuta Absoluta* insect monitored, forecasted and detected at an early stage by designing a novel intelligent trap in which artificial intelligence is add for this purpose.

All previous research focused on detecting the *Tuta Absolute* pest through infected leaves and using methods that do not give results that meet the expectations. In addition, the information obtained is insufficient to limit or reduce the spread of the*Tuta Absolute* pest, and therefore there was an urgent need to design an intelligent trap that helps provide accurate information about the numbers of adults present and spread in places where tomato crops are grown. This research proposes a novel intelligent trap, integrated with artificial intelligence, for early monitoring, forecasting, and detection of the *Tuta Absoluta* insect.

3. Research methodology

In this research an integrated system proposed for monitoring and controlling the spread of the *Tuta Absoluta* pest as shown in Fig. 3. A six sticky pieces used to catches the *Tuta Absoluta*, and the priority of these sticky pieces changes depends on human intervention or by setting a timer, that changes the sticky pieces periodically. The sticky piece goes to a place prepared to take clear images, which in turn goes to the trained artificial intelligence network that downloaded on the Jetson Nano developer kit to help makes the right decisions to reduce the spread of the *Tuta absoluta* pest. The intelligent system consists of two main parts; each part is equipped separately from the other and integrated into one system. The first part is responsible for designing the trap and its electronics and mechanism of action, while the other part is based mainly on the use of artificial intelligence and its correct placement in order to work with the trap with an integrated system so that it can early detection and control the spread of the *Tuta Absoluta* pest.



Fig. 2. *Tuta Absoluta* affected tomato crops.

Table 2
Datasets collects through different methods.

Experiments	No. of samples collected
Water traps	1668 samples
Sticky traps	5383 samples
others	115 samples

3.1. Dataset

In order to add artificial intelligence and obtain high accuracy during the training process in monitoring and following up the spread of insect pests, it is necessary to provide clear and accurate images in order to facilitate the training process. In addition, because insects are moving and flying, it was difficult to obtain many different images with good clarity. To address this problem, a group of researchers turned to the process of direct pinning the insect in the laboratory in order to take clear pictures in ideal laboratory conditions as shown in Fig. 4. In order to add artificial intelligence in the process of controlling the insect pest, the greater the number of different images in the training process, the better it will be in obtaining results, and the fact that these insects are in ideal conditions, they can be used in a few and limited applications. Real samples must take from the fields. Many practical methods used traps to catch insect pests and used various means to attract insects such as color [46,47], light [48], insect pheromones (sticky or water) traps and others [49,50].

In this research, the dataset used in training collected manually, and one of the most important effective methods is the use of traditional traps as shown in Fig. 5(a). The traps attract the males of*Tuta Absoluta* pest using sex pheromone, and after the *Tuta Absoluta* pest caught as shown in Fig. 5(b), the sticky base be taken to a suitable place to get images in a suitable environment by controlling the lighting and angle to take images such as shown in Fig. 5(c). The number of *Tuta Absoluta* collected with different images using the traditional sticky traps more than (5383) samples as mentioned in Table 2. Moreover and In order for the dataset to be good and meet the ambitions, water traps were set to catch the largest possible amount of the *Tuta Absoluta* pest and collect the required images by taking many different cases, as shown in Fig. 5(d) the use of water traps. The collected Samples through water traps in different images more than (1668) samples as mentioned in Table 2. Fig. 5 shows the use of a traditional trap and water trap in the process of collecting images.

A total of 320 different images collected, and since the AI methods used as objects detections, for that reason there are many cases in

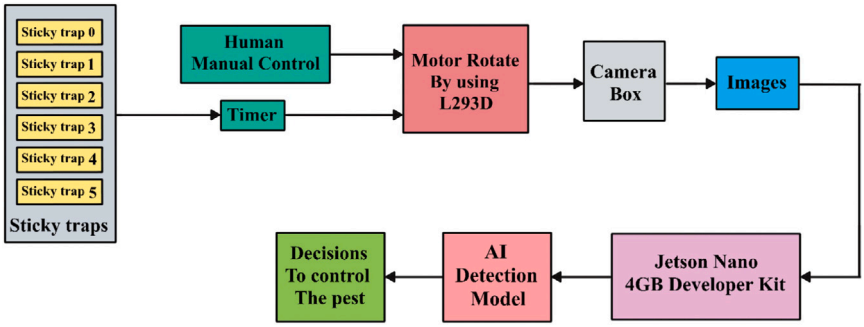


Fig. 3. Block diagram of the proposed system.



Fig. 4. Limited images can take for mummification insects.



(a) Traditional trap with pheromone to catch pest

(b) sticky traps

(c) collects images with suitable environment



(d) water traps

Fig. 5. Collects images for dataset.

the images. Some collected images have more than 1600 samples in one image and other images contain one sample in the image, as in Table 2. The images collected with different resolutions. The images collected using a microscope in scientific laboratories and collected at a resolution of 4608 × 3456, 5184 × 3456, and 3456 × 2592. The images collected for traditional sticky traps collected at a resolution of 4000 × 3000 and 4608 × 3456, the images collected for water traps collected at a resolution of 4000 × 3000, 2592 × 1942 and at 1200 × 1600 resolution, and there are other images collected at different resolutions.

3.2. Intelligent electronic trap

The use of traps to control insects began a very long time ago, and traps continued to develop continuously in order to reduce the risk of insect pests. Insects differ among themselves in characteristics, so the nature of the insect must studied and traps added to monitor it in order to obtain the most accurate information. The great development in technology has facilitated human life, so in this research, the design of the intelligent trap was relied on technology to reduce effort and time and to obtain information with higher accuracy. The trap designed to operate automatically to provide the information required to monitor the spread of the *Tuta Absoluta* pest, as shown in Fig. 6.

The trap consists of the upper hierarchical part, which is similar in design to traditional traps, and this inventory is fixed and immovable, and as in Fig. 7. Several field experiments were conducted to catch *Tuta Absoluta* insects when collecting datasets, and traps were used in different colors such as (yellow, red, blue, white, transparence, black and green), and the insects were caught as shown in Fig. 8. After continuous follow-up, it found that the color of the trap has no significant effect on the process of attracting insects, as it is possible to use red, yellow, green, white or blue in the design of this modern trap.

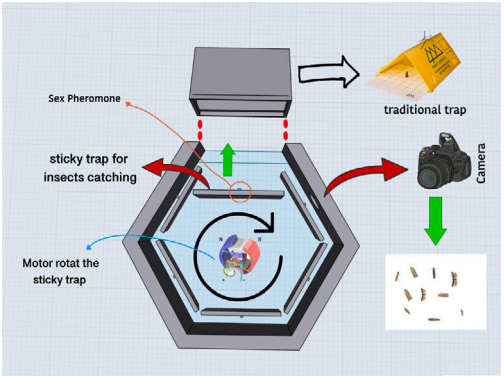
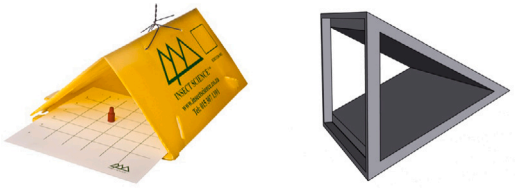


Fig. 6. The design of proposed novel trap.



(a) traditional trap (b) proposed trap

Fig. 7. The upper hierarchical part used in traditional trap and the proposed trap.

Table 3
The concentration of the pheromone used with average Insects catches per week.

Experiments	Exp.1	Exp.2	Exp.3	Exp.4	Exp.5	Exp.6
Concentration	0.5 µg	1 µg	2 µg	3 µg	4 µg	5 µg
Av. Insects catch per week	125	175	119	122	170	200

The novel trap contains a sticky substance similar to the traditional trap in order to help catch insects. Each sticky trap has a sex pheromone substance, which attracts adult males to the trap, as shown in Fig. 9 which shows the sticky trap after and before catching insects with the presence of sex pheromone on top of the adhesive. The concentration of the pheromone used is especially important in the different numbers of males attracted to the trap. It is important to study the effect of concentrations in order to determine the optimal concentration and according to the environment so that the number caught is more accurate and realistic. In addition, it is important to find out the minimum and maximum concentrations used within the specified space of the greenhouses without the effect of dispersal attraction. Table 3 shows the concentration of the pheromone used with average insects catches per week.

There is an effect in choosing the appropriate height of traps in catching adults. It is important to determine the height of pheromone traps, whether in greenhouses or in fields, because of this effect on the accuracy of determining the numbers of caught males with the effectiveness of their attraction to traps, Table 4 shows the results of the effect of the height of trap to catch insects of *Tuta Absoluta*. The trap contains a container in the form of a box, which is the basis for controlling the follow-up of the nature of the designed trap and contains six sticky traps. And the container synchronizes the movement of the sticky traps in order to facilitate the process of catching insects periodically and the process of taking pictures of the sticky trap when the process of attracting and catching insects that occurs periodically is completed such as shown in Fig. 10.

The motor in the container used in the rotation process that controls the synchronization of the presence of the sticky trap to catch insects and at the same time to synchronize taking images periodically through the digital camera within the design of the trap. The rotation of dc

Table 4
The effect of height in the catch the adults of *Tuta Absoluta*.

Height (cm)	Average per week
50	60.0
100	49.0
150	35.0
200	16.5
300	8.5

motor controlled by sending exact signals to L293D IC in order to rotate the motor with exact rotation and direction, Fig. 11 shows the L293D IC rotation mechanism. There will be two ways to control the motor. The first one is manual, by sending direct signals to the motor to control its rotation and change the traps to detect the insects. The second way is by setting a specific time, based on which a signal will be sent to the motor to rotate at time intervals. These intervals are determined by people specialized in agriculture as needed.

One of the most important points was take into account the design of the proposed intelligent trap was simplicity in design and ease of use. The intelligent trap placed in the place where tomato crops are grown. The proposed trap attracts males via the sex pheromone to the sticky piece, and after a specific period of time according to Whether it is a day, a week, a month, etc., The intelligent trap replace the adhesive piece with a new one and transfer the used adhesive piece to a small box, which considered a suitable environment for taking a new picture. After that comes the important role of artificial intelligence. The captured image enters the pre-trained network, which in turn carries out the process of detecting and identifying the *Tuta Absoluta* pest in the adhesive piece by placing a box with a title above each box, detecting the *Tuta Absoluta* pest with very high accuracy.

3.3. Object detection with deep learning

Classification methods was very important in recent years, it takes a large part of attention of researchers, but it is important to identify the object and identify it by surrounding the detected object with a box and title that distinguishes the objects in a way that differs from the classification of images in order to provide information with higher accuracy. This task is referred to as object detection [51]. The state of the art detectors are deep neural networks, namely convolution neural networks (CNN). Redmon et al. designed You Only Look Once (YOLO) framework using the topmost feature map for localization and classification [52]. It was continued to be developed into YOLOv2 and YOLOv3 [53] and they stopped to this point in order to come after him, Alexey Bochkovskiy et al. to develops this algorithm into YOLOV4 and until it reached the stage of development to YOLOv5 [54]. The development of YOLO continued through YOLOV6 [55], YOLOV7 [56], YOLOv8 [57], which is focuses on the box size that surrounds the object and continue in developing new virsion of YOLOv9 [58], in 2024 YOLOv10 [59] produced until it reached to the last version of YOLOv11 [60].

3.3.1. Performance evaluation

Object detection is one of the most important challenges facing researchers at the present time and for many problems where a square or any other shape is drawn to detect the object in the image, so it is necessary to use the appropriate measurements to determine the exact target because the sizes of the objects are different in most times, And it is necessary to evaluate the performance of an object detector, and the most common Eqs. (1) to (3) are used for this purpose.

$$precision = \frac{T_p}{T_p + F_p} \tag{1}$$

$$recall = \frac{T_p}{T_p + F_n} \tag{2}$$



(a) red trap (b) green trap (c) white trap

Fig. 8. Different color of trap used to test which color are more attractive to the Tuta Absoluta insects.

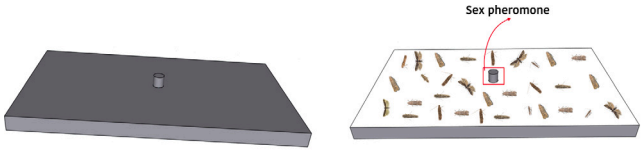


Fig. 9. The sticky component with pheromone from the proposal.

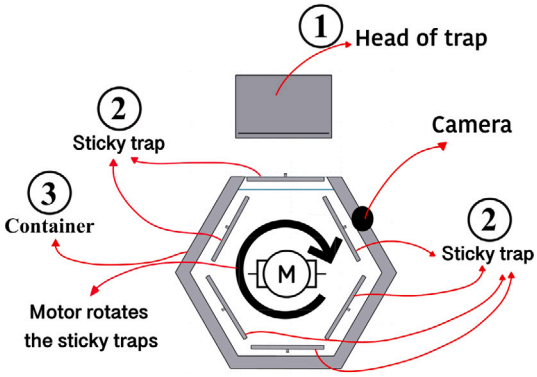


Fig. 10. Schematic diagram of automated rotating sticky trap system.

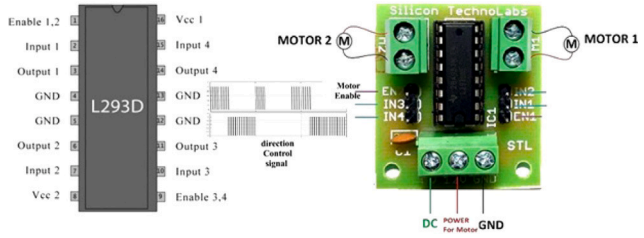


Fig. 11. L293D IC.

$$m_{AP} = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{3}$$

True positive (TP) and False positive (FP) is a correct and incorrect detection of an object that actually exists in the picture respectively, (FP) means that the network marks an object that is not there in the picture. False negative (FN) means that there is an object but the network does not discover it. The intersection over union (IoU) Calculates the intersection in the region between the ground truth bounding box and predicted bounding box, And by Comparing the IoU with a given threshold, detection can be classified as incorrect or correct. Each value of the IoU threshold provides a different average precision (AP) metric, so it is necessary to specify this value.

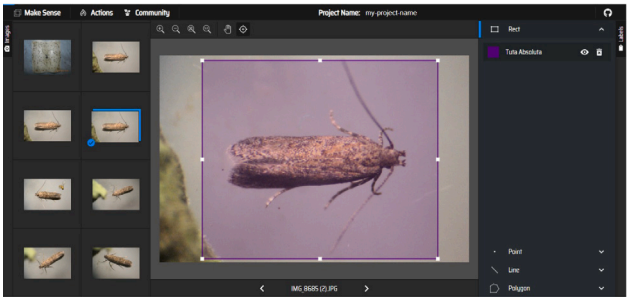


Fig. 12. Annotations of the dataset.

Table 5
The results for different experiments.

Experiments	Epoch	Precision	Recall	mAp 0.5	mAp 0.5:0.95
70:20:10	256	80.9	75.6	84.2	47.9
60:20:20	379	83.6	93.3	90.6	57.2
62.5:25:12.5	323	83.3	87.7	88.1	54.2
76.5:13.5:10	328	69.3	93.2	85	51.6

4. Dataset preparation and annotation

The diversity of collect the images is necessary to increase the efficiency of using artificial intelligence during the training process. In order for the images included in the training process to be prepared for the purposes of detecting objects, it is necessary to annotate all the images included in the training process, where a box surrounds the objects to detect and directly above it is the labels of the objects such as shown in Fig. 12. Moreover, as in the following images in Fig. 13, (YOLOv11, YOLOv9, YOLOv8 and YOLOv5), which used in the training process, where the number of images increased after cutting part of the images and merging them with other parts cut from other images in order to generate new images added to the dataset used.

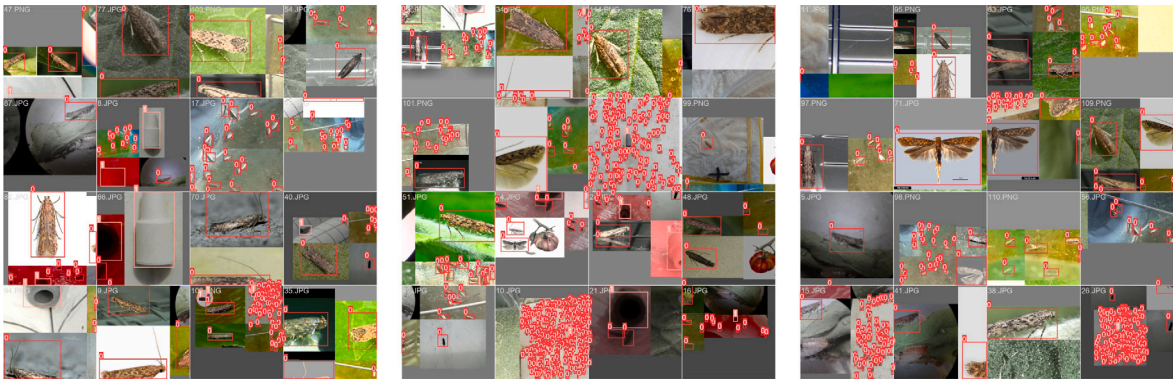
5. Experimental study

This section discusses the experiments conducted during this research, the results obtained and the mechanism for choosing the model in the proposed system.

5.1. Experimental settings

In order to train the artificial intelligence, it is necessary to divide the previously collected dataset images into three partitions (training set, validation set and test set). In many previous researches, the percentages of distribution of the number of images in the files were different and random with different percentages. In this paper, many experiments are conducted based on the ratios used in previous research, to reach the best distribution and to reach good results such as those shown in Table 5.

In the first experiment, the images were distributed in the three partitions in a ratio of (70:20:10), respectively, as used [4] And the training procedure consisted of 265 epochs, which required time (0.718 h)



(a) samples of dataset used (b) samples of dataset used (c) samples of dataset used

Fig. 13. Examples of the samples used to train the models.

Table 6
Summary about the top results per model for YOLOv5.

Model	Epoch	Precision	Recall	mAp 0.5	mAp 0.5:0.95	F1
YOLOv5m	400	92.3	80.7	91.4	57	86
YOLOv5s	379	83.6	93.3	90.6	57.2	88
YOLOv5n	314	88.8	59.7	68.3	37	69
YOLOv5x	292	95.9	92.5	94	61.6	94

Table 7
Summary about the top results per model for YOLOv8.

Model	Epoch	Precision	Recall	mAp 0.5	mAp 0.5:0.95	F1
YOLOv8n	219	54.8	67.5	66.6	34.3	64
YOLOv8s	117	57.8	72.1	61.8	27.6	65
YOLOv8m	93	80.7	52.0	65.1	33.4	56
YOLOv8l	204	87.1	89.8	90.9	47.0	88
YOLOv8x	216	89.3	75.1	82.5	45.7	82

Table 8
Summary about the top results per model for YOLOv9.

Model	Epoch	Precision	Recall	mAp 0.5	mAp 0.5:0.95	F1
gelan-c	185	92	73.2	80.8	48	82
gelan-e	109	88.8	76.1	83.7	45.4	81
YOLOv9-c	258	87.1	93.5	92.6	53.9	91
YOLOv9-e	186	89.5	90.1	90.7	56.3	90

Table 9
Summary about the top results per model for YOLOv11.

Model	Epoch	Precision	Recall	mAp 0.5	mAp 0.5:0.95	F1
YOLOv11L	210	76.4	78.1	80.8	45.7	77
YOLOv11m	163	84.3	86.4	84.1	46.9	86
YOLOv11n	232	71.4	79.2	75.4	41.7	75
YOLOv11s	207	90.8	70.7	78.4	37.7	79

to train the dataset; the training results were precision 80.9% and recall 75.6%. In the second experiment, the images were distributed in the three partitions in a ratio of (60:20:20), respectively, as in [24] And the training procedure consisted of 379 epochs, which required time (1.026 h) to train the data set; the training results were precision 83.6% and recall 93.6%. In the third experiment, the images were distributed over the three partitions at a ratio of (62.5:25:12.5) as used [30] And the training procedure consisted of 332 epochs, which required time (0.856 h) to train the data set; the training results were precision 83.3% and recall 87.7%. In the fourth experiment, the images were distributed over the three partitions in a ratio of (76.5:13.5:10), as used [43] and the training procedure consisted of 328 epochs, which required time (0.997 h) to train the data set, the training results were precision 69.3% and recall 93.2%. It is clear and after comparing all the results obtained from the four experiments, the second experiment is considered the best for obtaining good results in the training process, in which the images were distributed to the three files in percentages (train 60%: validation 20%: test 20%).

5.2. Results and comparison of models

In order for the process of using the trained network in the proposed system, it is very necessary to compare the results obtained with the other versions of YOLOv5, YOLOv8, YOLOv9 and YOLOv11, The same dataset used to train all the models, the comparative results on several key performance indicators presented in Table 6, Table 7, Tables 8 and 9 respectively.

There are an infinite number of bounding boxes that should not detected within any given image, and for that reason a true negative (TN) result does not use in objects detection algorithms. This makes it impossible to use any matrices that depend on TN [61]. In terms of speed and accuracy, the models are very similar and slightly different, the models tested on the same datasets, the differences between the models in terms of precision, recall, mAP 0.5 and mAP 0.5:0.9. It is clear from the tables and models graphs that the results obtained are at their best when YOLOv5x is used, as the precision value equal to 95.9% was obtained, recall equal to 92. 5% and the values of mAP 0.5 are equal to 94%. Then the results are better than others using YOLOv9-c, YOLOv9-e, and YOLOv8L respectively; then comes the rest of the YOLO5, YOLOv8, YOLOv9, and YOLOv11 models in terms of the possibility of using them, because no results obtained were as ambitious as the results obtained by relying on YOLOv5x. If the results obtained are compared through the F1 score values, it is again clear that the values obtained through YOLOv5x are the best compared to the rest of the modes that were used for the training process, as they obtained an F1 score value equal to 94%. The lowest value of the F1 score was obtained by YOLOv8 m and the value was equal to 56%.

The models graphs obviously continues to learn throughout all epochs because performance results keep increasing, and each of these models needed epochs different from the rest of the models to get the desired results from training. In Fig. 14 it represents the precision curves for the different YOLOv5 and YOLOv9 models, it is clear that the values started with small values and quickly began to increase through the training process with many differences during each epoch. In addition, it is clear that the results of YOLOv5x are more stable and more accurate if compared to the rest of the other models, it is noted that YOLOv5 m and gelan-c are good and stable, but they are not the same as YOLOv5x. The least stable result achieved using YOLOv8n. The best results obtained in terms of stability and accuracy for the YOLOv8x YOLOv11 models are good, but they are not better than the results obtained using the YOLOv5x and YOLOv9 models.

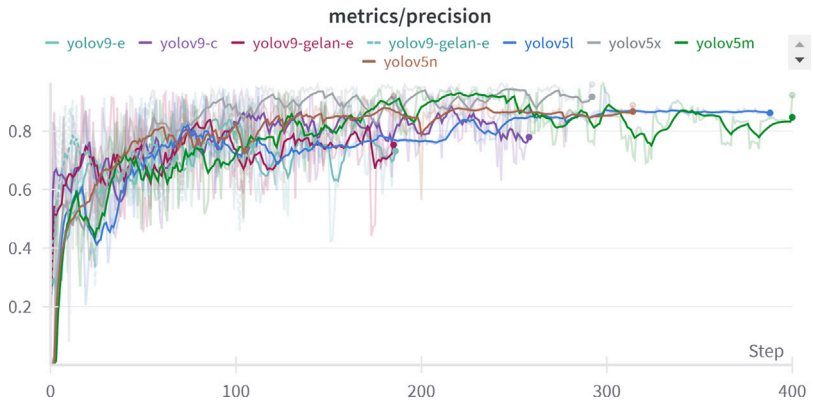


Fig. 14. The precision metric results using YOLOv5 and YOLOv9 models.

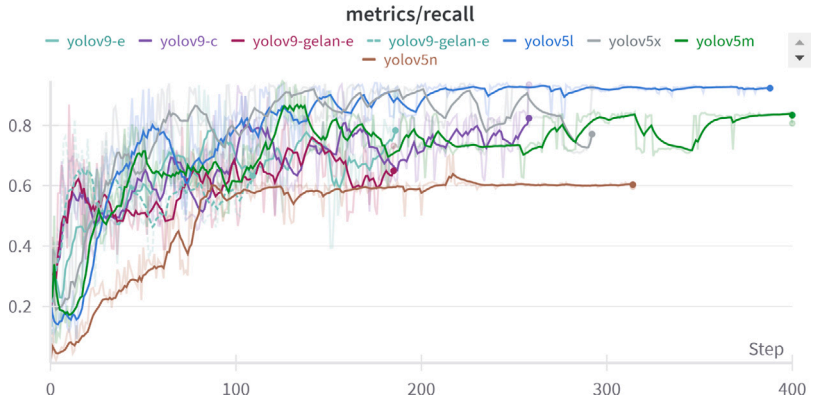


Fig. 15. The recall metrics results using YOLOv5 and YOLOv9 models.

In Figs. 15 the recall curves shown by using YOLOv5 and YOLOv9 models. The recall was considered one of the most important performance that characterizes the accuracy of the trained network in object detection algorithms. In Fig. 15, it seems clear that all the curves started with small values and rapidly increased during the training process. It is noticeable that the YOLOv9-c curve tends to be more stable than other curves. The closest to YOLOv9-c is the result obtained using YOLOv5s, YOLOv5x and YOLOv9-e respectively, and the least in terms of instability was the result obtained using YOLOv8m and YOLOV5n respectively. In the proposed system, the rest of the performance must be taken into account to ensure that the trained network is accurate and can be relied upon. And for the purpose of comparison, YOLOv8 and YOLOv11 models used, as it is clear that the results obtained are not as good as those obtained using YOLOv5 and YOLOv9, and all the curves had their values change and increase during the epochs, but they did not meet the ambition.

To ensure that the trained network is more effective, it is better to find the area under the curve, which is the relationship between precision and recall, because the values of precision and recall range from zero to one, and because the zigzag relationship between the values of precision and recall. Therefore, 101 interpolation used to make the relationship smooth, and through it, the values of the area under the curve calculated, which is in object detection methods is mAP 0.5 [61]. From Figs. 16 and Tables above it is clear that the mAP 0.5 values obtained using YOLOv5x are the best and most accurate in detecting objects as compared to all (YOLOv5, YOLOv8, YOLOv9 and YOLOv11 respectively) models used, and YOLOv9-c comes after it in terms of stability and obtaining the second highest values of mAP 0.5 And then YOLOv8l, YOLOv9-e and YOLOv5s respectively. YOLOv8 and YOLOv11 models used for comparison purposes were at their best when using (YOLOv8l and YOLOv11m) models; but it is less than the results obtained using YOLOv5x and YOLOv9-c. The lowest values were obtained while using (YOLOv8s, YOLOv8m, YOLOv8n and YOLOv5n) respectively during the training process and the least reliable because it was unstable and did not give satisfactory and convincing results. The mAP 0.5 values started with small values and increased during the training, this means that the trained network learned correctly during each epoch. However, there is a difference in stability when obtaining mAP 0.5 values, as shown in Fig. 16.

To ensure that the model used is efficient and meets the ambition, measuring and comparing the results obtained to mAP 0.5:0.95; as it calculated through several special IOU values, which start with a value of 0.5 until the last value, which is 0.95, with an increasing rate of IOU values with a value of 0.05. In Fig. 17 and tables above, it is also clear that the results obtained using YOLOv5x are the best, while the model that obtained the least convincing results was YOLOv8s, The closest results during the training process were YOLOV5s, YOLOv5m, YOLOv9-e and YOLOv9-c . The rest of the models, especially those used in YOLOv8 and YOLOv11, had different values, and YOLOv8l was the best among its peers, but it was not the best when compared to YOLOv5x.

The loss function is important because the smaller the loss, the better the classifier is at modeling the relationship between the output targets and the input data, and it is used to show the performance of a given predictor in classifying the input data points in the dataset used. Fig. 18, it is an indicator of the accuracy of the trained model in determining the bounding box. the lower the values the better, as it is an indicator of the accuracy of determining the center, the height and width of the bounding box in detecting the object in the image. At the beginning and during the training process, the values are large, and during the epochs, the values decrease. If the values are rising, this indicates that there is a defect during the training process. In Fig. 18, for

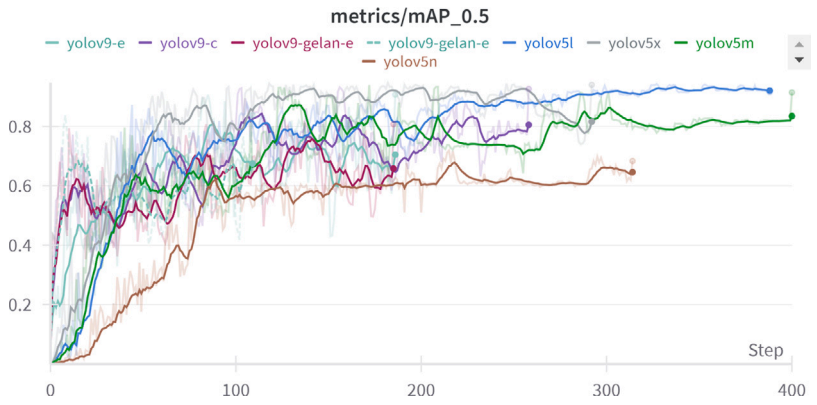


Fig. 16. The mAP:0.5 metric results using YOLOv5 and YOLOv9 models.

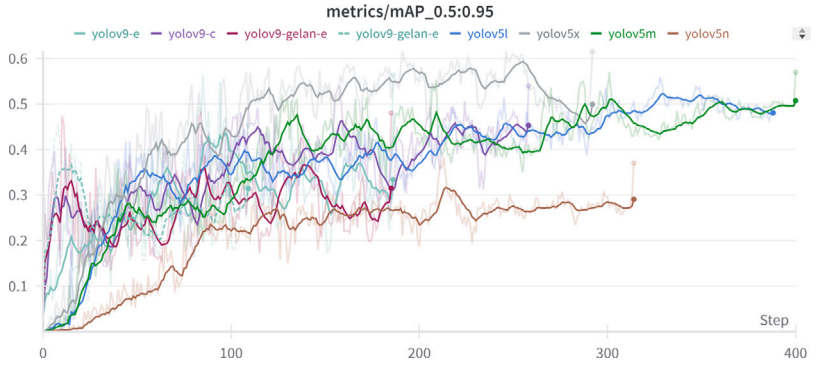


Fig. 17. The mAP:0.5–0.95 metric results using YOLOv5 and YOLOv9 models.

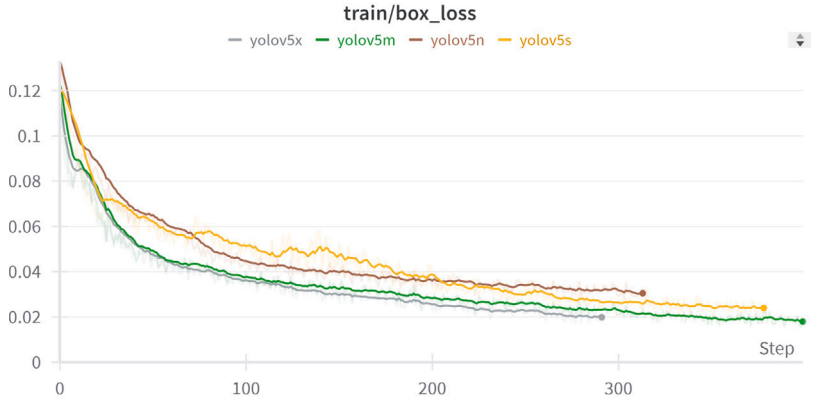


Fig. 18. Box loss using YOLOv5 models.

all models used based on YOLOv5, it is noted that the box loss values decrease with each training process, and these values are considered much better if compared with the results obtained during the training process using (yolov11, YOLOv9 and YOLOv8) models.

In Fig. 19 represents classes loss, which represents the standard that indicates losses by classifying objects into correct classes and is considered very important and helpful in improving the performance of the model used. It noted that using YOLOv5 models, all values after starting the training process began to decrease, and this indicates that the models started learning correctly. It is clear that the results obtained by using YOLOv5 models are better when compared to (YOLOv11, YOLOv9 and YOLOv8) models, as the class loss values decrease with training steps go through epochs, but they do not meet the ambition as shown in the results That was relied on YOLOv5.

In Figs. 20, it represents the losses that measure the accuracy of the model in determining the presence or absence of an object within the bounding box in the image. The smaller losses, the better reliability of the model used. It is clear that at the beginning of the training process using YOLOv5 models, the loss values were large and quickly began to decrease during the training steps. This indicates that these models trained correctly, and if it is the opposite, this indicates that the models cannot relied upon, due to the possibility that the objects detected incorrectly. The results of (YOLOv11, YOLOv9 and YOLOv8) models not better than the values obtained during the training process relying on YOLOv5.

5.3. Prediction examples

Usually, as mentioned previously, the model tested after completing the training process, using other images that not used for training the model. Fig. 21 are an example of the results of a trained model test

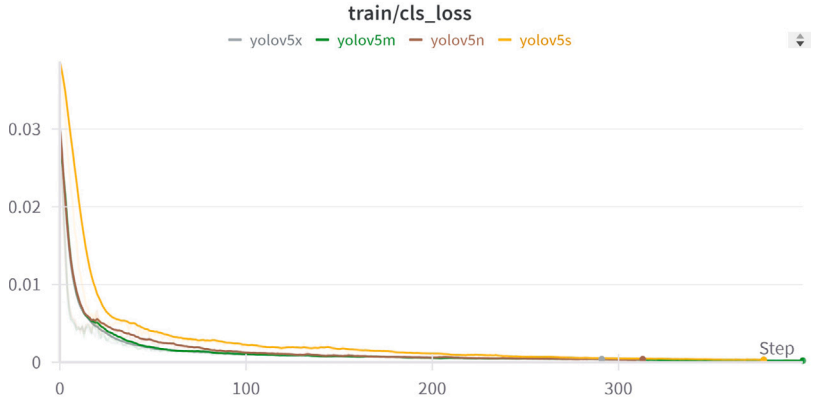


Fig. 19. Classes loss using YOLOv5 models.

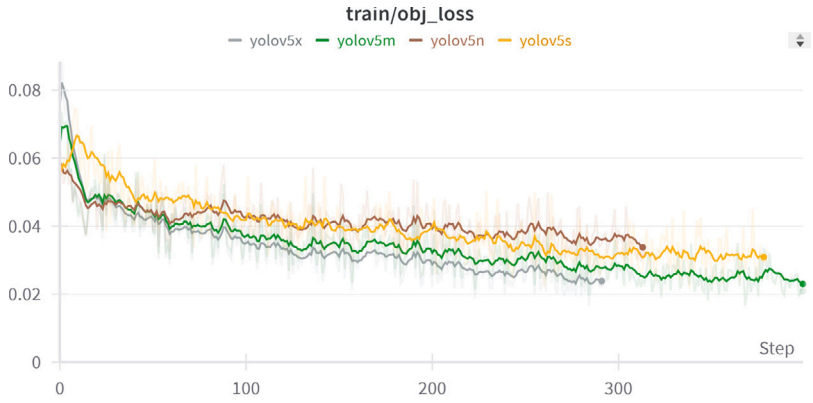


Fig. 20. Object loss using YOLOv5 models.

where objects bounded by a bounding box and each bounding box has an address above the box by which the object detected.

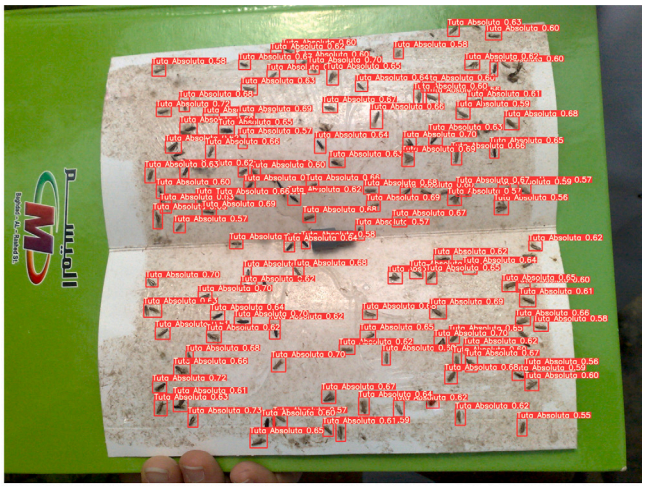
Before conducting the test, the value of the IoU parameter was lowered to 0.4, which indicates that the detection box is valid in the case of IoU≤40%. This value was taken after conducting several tests and according [52–54] where there is difficulty in identifying or an error occurs in diagnosing the type of insects being a small object in order to avoid the wrong diagnosis in identifying the objects in the picture.

6. Discussion

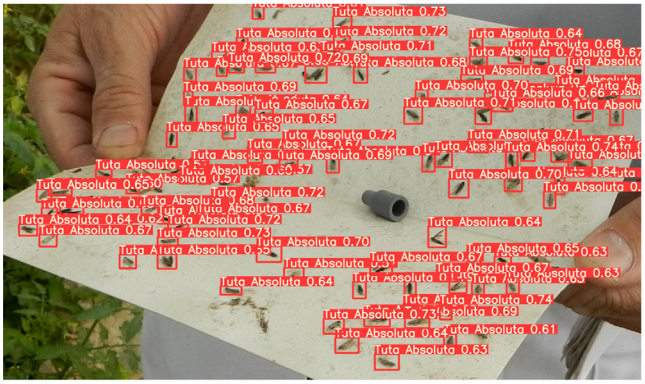
The use of YOLOv5x to detect the *Tuta Absoluta* pest and to help calculate the number of male insects caught in the trap, because knowing the number of *Tuta Absoluta* insects facilitates the process of controlling the spread of the pest. After conducting many experiments in releasing different numbers of *Tuta Absoluta*, in several experiments, it was called "release and recapture", Where a number of Tuta Absoluta insects released and during different periods the number of insects caught in the trap was monitored and it was concluded that it was possible to know the true approximate numbers of *Tuta Absoluta* insects in tomato crops based on the number of insects caught in the trap. The extent of crop infestation been classified into three levels:

- In the event that the trap catches a number of adult *Tuta Absoluta* insects with low number of insects during one week, it considered in the low-risk stage.
- In the event that the trap catches a number of adult *Tuta Absoluta* insects with medium number of insects during one week, it considered to be in the medium risk stage.
- In the event that the trap catches a number of adult *Tuta Absoluta* insects with high than normal insects during one week, it considered in the high-risk stage.

All this information provided by the artificial intelligence that has been trained is important in the process of controlling the spread of the *Tuta Absoluta* pest and monitoring it on an ongoing basis, but it is not the only information that is required in conducting human intervention for the purpose of combating the pest,when sufficient information obtained, the specialist in the agricultural field will instructed to monitor and release parasites or predators in the field and calculate the effect that occurs continuously through artificial intelligence. There are many different parasites and predators all over the world, including as in [62–74]. The larval parasitoids are among the biological agents that have been used in many regions of the world to control some agricultural pests, especially with lepidoptera, and most of these parasitoids belong to the Braconidae family, which is one of the largest parasitic families and the most used in life resistance programs, as it includes parasitoids whose members rarely take the nature of secondary parasitism, It includes many subfamilies, including Braconinae, and many genera follow to it. They are external parasites that are often collective on scaly-winged larvae. They are larval parasitoids that parasitize mainly on lepidoptera, coleoptera and larvae. The host larvae anesthetized with toxic secretions from the intruder, paralyzing them, and during its lifetime, the female parasite B. brevicornis lays many eggs on the larvae. In this research, the parasitoid B. brevicornis was proposed to use because it is one of the ectoparasites that parasitizes on scaly-winged larvae with high efficiency [75] and because of its positive aspects that support its adoption in biological control operations because it has a short life span and has a wide family range and high productivity. It is very likely that the number of parasitic insects present in tomato crops



(a)



(b)

Fig. 21. a and b are examples of detection of an objects using test image, *Tuta Absoluta* bounding with red box and identified with label above the box.

will be sufficient to deal with the continuous spread of the *Tuta Absoluta* blight without spraying chemical pesticides on crops to avoid exposing human life to health and environmental risks. Therefore, monitoring continues on an ongoing basis to follow the spread of the *Tuta Absoluta* pest and the numbers of parasitic insects for *Tuta Absoluta*, and in the event that the parasitic insects are unable to treat the spread of the *Tuta Absoluta* pest, information is sent for the purpose of human intervention by releasing parasites that were raised in cages for the purpose of control. If the pest cannot controlled by parasites, it will be done the last resort is to use insecticides at reasonable rates according to the presence of pest density in the crops.

In the case of relying on other predators or parasites to limit and control the spread of the pest, the required statistics should studied in real time, which depend on environments changes. The lack of information is difficult to obtain due to the lack of intelligent devices that help collect samples and images of Tuta absoluta pests. However, a solution to this problem has been proposed, providing more information by employing the proposed system that facilitates this process.

After performing all these steps, through which modern novel intelligent traps designed and used, it became possible to answer the questions that previously posed. Where it is possible to rely on the modern designed intelligent trap and with higher efficiency than the traditional traps that used previously. It is not obligatory to control the pest by spraying with chemical pesticides, as it is possible to rely on the natural treatment provided by the environment to reduce various health risks.

7. Conclusions

Providing healthy food in large quantities is an urgent necessity in the present time, and reducing dependence on pesticides to solve pest problems in agriculture is a great challenge. Because the use of pesticides to get rid of pests and insects has very significant effects on human health, including many different diseases. Therefore, in this research, an intelligent electronic trap designed that contributes directly to controlling the spread of pests and identifying the pest with very high accuracy, especially in tomato crops. Technology has become an integral part in our daily lives and has proven its high efficiency in many applications in recent years, therefore, artificial intelligence was represented by relying on YOLOv5x for the purpose of object detection, as it contributes to identifying *Tuta Absoluta* and sending information in a continuous manner that directly helps in making the optimal decision. The designed trap controlled by L293D IC, which can control the time in which the traps change and rotate as needed. In a timely manner to reduce the risk of the spread of the pest, and to stay away as much as possible from the use of pesticides in the pest control process to reduce health risks. Several models tested to determine the best method for use with the intelligent electronic trap. The results varied depending on the method chosen. Comparisons made with (YOLOv11, YOLOv9, YOLOv8, and YOLOv5). The best results, and those that met the expectations for use with the intelligent electronic trap, achieved when YOLOv5x used. The results of artificial intelligence training were excellent depending on the dataset collected personally, where the precision was 95.9%, recall was 92.5%, mAP 0.5 was 94% and F1 score was 94%. In addition to the simple design of the trap made it easy to use, low cost and reliable in the fields of modern agriculture.

CRediT authorship contribution statement

Yaser M. Abid Alasady: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eduardo Pérez:** Visualization, Resources, Writing – review & editing. **Sebastián Ventura:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Funding acquisition, Data curation, Conceptualization.

Financial disclosure

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported in part by the PID2020-115832GB-I00 and TED2021-132702B-C22 projects of Spanish Ministry of Science and Innovation and the European Regional Development Fund, and by the ProyExcel-0069 project of the Andalusian University, Research and Innovation Department.

Data availability

Dataset will be made available on request. and the dataset are available at the link: alasady-tuta-absoluta-dataset.

References

[1] J.P.L. Escola, R.C. Guido, I.N. da Silva, A.M. Cardoso, D.H.B. Maccagnan, A.K. Dezotti, Automated acoustic detection of a cicadid pest in coffee plantations, Comput. Electron. Agric. 169 (2020) 105215.

[2] L. Mkonyi, D. Rubanga, M. Richard, N. Zekeya, S. Sawahiko, B. Maiseli, D. Machuve, Early identification of Tuta absoluta in tomato plants using deep learning, Sci. Afr. 10 (2020) e00590.

[3] R. Gandhi, S. Nimbalkar, N. Yelamanchili, S. Ponkshe, Plant disease detection using CNNs and GANs as an augmentative approach, in: 2018 IEEE International Conference on Innovative Research and Development, ICIRD, IEEE, 2018, pp. 1–5.

[4] L. Jiao, S. Dong, S. Zhang, C. Xie, H. Wang, AF-RCNN: An anchor-free convolutional neural network for multi-categories agricultural pest detection, Comput. Electron. Agric. 174 (2020) 105522.

[5] A. Sabarwal, K. Kumar, R.P. Singh, Hazardous effects of chemical pesticides on human health–Cancer and other associated disorders, Environ Toxicol Pharmacol 63 (2018) 103–114.

[6] S. Poudel, A. Dhakal, Integrated pest management (ipm) and its application in rice–a review, Rev. Food Agric. 1 (2) (2020).

[7] S.K. Dara, The new integrated pest management paradigm for the modern age, J. Integr. Pest Manag. 10 (1) (2019) 12.

[8] M.Z. Alam, A.R. Crump, M.M. Haque, M.S. Islam, E. Hossain, S.B. Hasan, S.B. Hasan, M.S. Hossain, Effects of integrated pest management on pest damage and yield components in a rice agro-ecosystem in the Barisal region of Bangladesh, Front. Environ. Sci. 4 (2016) 22.

[9] H. Wang, X. Pan, Y. Zhu, S. Li, R. Zhu, Maize leaf disease recognition based on TC-MRSN model in sustainable agriculture, Comput. Electron. Agric. 221 (2024) 108915.

[10] S. Lim, S. Kim, S. Park, D. Kim, Development of application for forest insect classification using CNN, in: 2018 15th International Conference on Control, Automation, Robotics and Vision, ICARCV, IEEE, 2018, pp. 1128–1131.

[11] X. Cheng, Y. Zhang, Y. Chen, Y. Wu, Y. Yue, Pest identification via deep residual learning in complex background, Comput. Electron. Agric. 141 (2017) 351–356.

[12] M. Ariza-Sentís, S. Vélez, R. Martínez-Peña, H. Baja, J. Valente, Object detection and tracking in precision farming: a systematic review, Comput. Electron. Agric. 219 (2024) 108757.

[13] D.I. Patrício, R. Rieder, Computer vision and artificial intelligence in precision agriculture for grain crops: A systematic review, Comput. Electron. Agric. 153 (2018) 69–81.

[14] J.G.A. Barbedo, Detection of nutrition deficiencies in plants using proximal images and machine learning: A review, Comput. Electron. Agric. 162 (2019) 482–492.

[15] T.D.C. Júnior, R. Rieder, Automatic identification of insects from digital images: A survey, Comput. Electron. Agric. 178 (2020) 105784.

[16] W.P. Amorim, E.C. Tetila, H. Pistori, J.P. Papa, Semi-supervised learning with convolutional neural networks for UAV images automatic recognition, Comput. Electron. Agric. 164 (2019) 104932.

[17] E.C. Tetila, B.B. Machado, G. Astolfi, N.A. de Souza Belete, W.P. Amorim, A.R. Roel, H. Pistori, Detection and classification of soybean pests using deep learning with UAV images, Comput. Electron. Agric. 179 (2020) 105836.

[18] Y. Li, H. Wang, L.M. Dang, A. Sadeghi-Niaraki, H. Moon, Crop pest recognition in natural scenes using convolutional neural networks, Comput. Electron. Agric. 169 (2020) 105174.

[19] N. Liang, S. Sun, J. Yu, M.F. Taha, Y. He, Z. Qiu, Novel segmentation method and measurement system for various grains with complex touching, Comput. Electron. Agric. 202 (2022) 107351.

[20] H. Yang, F. Chang, Y. Huang, M. Xu, Y. Zhao, L. Ma, H. Su, Multi-object tracking using deep SORT and modified CenterNet in cotton seedling counting, Comput. Electron. Agric. 202 (2022) 107339.

[21] N. Genze, R. Ajekwe, Z. Güreli, F. Haselbeck, M. Grieb, D.G. Grimm, Deep learning-based early weed segmentation using motion blurred UAV images of sorghum fields, Comput. Electron. Agric. 202 (2022) 107388.

[22] F. Wang, R. Wang, C. Xie, P. Yang, L. Liu, Fusing multi-scale context-aware information representation for automatic in-field pest detection and recognition, Comput. Electron. Agric. 169 (2020) 105222.

[23] Q.-J. Wang, S.-Y. Zhang, S.-F. Dong, G.-C. Zhang, J. Yang, R. Li, H.-Q. Wang, Pest24: A large-scale very small object data set of agricultural pests for multi-target detection, Comput. Electron. Agric. 175 (2020) 105585.

[24] A.S. Almryad, H. Kutucu, Automatic identification for field butterflies by convolutional neural networks, Eng. Sci. Technol. Int. J. 23 (1) (2020) 189–195.

[25] A. Cruz, Y. Ampatzidis, R. Pierro, A. Materazzi, A. Panattoni, L. De Bellis, A. Luvisi, Detection of grapevine yellows symptoms in Vitis vinifera L. with artificial intelligence, Comput. Electron. Agric. 157 (2019) 63–76.

[26] Y. Yue, X. Cheng, D. Zhang, Y. Wu, Y. Zhao, Y. Chen, G. Fan, Y. Zhang, Deep recursive super resolution network with Laplacian pyramid for better agricultural pest surveillance and detection, Comput. Electron. Agric. 150 (2018) 26–32.

[27] X. Hao, J. Jia, W. Gao, X. Guo, W. Zhang, L. Zheng, M. Wang, MFC-CNN: An automatic grading scheme for light stress levels of lettuce (Lactuca sativa L.) leaves, Comput. Electron. Agric. 179 (2020) 105847.

[28] C.R. Jensen, A. Battilani, F. Plauborg, G. Psarras, K. Chartzoulakis, F. Janowiak, R. Stikic, Z. Jovanovic, G. Li, X. Qi, et al., Deficit irrigation based on drought tolerance and root signalling in potatoes and tomatoes, Agricult. Water. Manag. 98 (3) (2010) 403–413.

[29] J.A. Monte, D.F.d. Carvalho, L.O. Medici, L.D. da Silva, C. Pimentel, Growth analysis and yield of tomato crop under different irrigation depths, Rev. Bras. Eng. Agrícola e Ambient. 17 (2013) 926–931.

[30] J. Shijie, J. Peiyi, H. Siping, et al., Automatic detection of tomato diseases and pests based on leaf images, in: 2017 Chinese Automation Congress, CAC, IEEE, 2017, pp. 2510–2537.

[31] Q. Wang, F. Qi, M. Sun, J. Qu, J. Xue, Identification of tomato disease types and detection of infected areas based on deep convolutional neural networks and object detection techniques, Comput. Intell. Neurosci. 2019 (2019).

[32] J. Liu, X. Wang, Early recognition of tomato gray leaf spot disease based on MobileNetv2-YOLOv3 model, Plant Methods 16 (2020) 1–16.

[33] X. Wang, J. Liu, X. Zhu, Early real-time detection algorithm of tomato diseases and pests in the natural environment, Plant Methods 17 (1) (2021) 43.

[34] M. Brahimi, K. Boukhalfa, A. Moussaoui, Deep learning for tomato diseases: classification and symptoms visualization, Appl. Artif. Intell. 31 (4) (2017) 299–315.

[35] K.P. Ferentinos, Deep learning models for plant disease detection and diagnosis, Comput. Electron. Agric. 145 (2018) 311–318.

[36] K. Zhang, Q. Wu, A. Liu, X. Meng, et al., Can deep learning identify tomato leaf disease? Adv. Multimed. 2018 (2018).

[37] J. Zhang, J. Xie, F. Zhang, J. Gao, C. Yang, C. Song, W. Rao, Y. Zhang, Greenhouse tomato detection and pose classification algorithm based on improved YOLOv5, Comput. Electron. Agric. 216 (2024) 108519.

[38] W. Chen, M. Liu, C. Zhao, X. Li, Y. Wang, MTD-YOLO: Multi-task deep convolutional neural network for cherry tomato fruit bunch maturity detection, Comput. Electron. Agric. 216 (2024) 108533.

[39] M. Jang, Y. Hwang, Tomato pose estimation using the association of tomato body and sepal, Comput. Electron. Agric. 221 (2024) 108961.

[40] Z. Yang, L. Sun, Z. Liu, J. Deng, L. Zhang, H. Huang, G. Zhou, Y. Hu, L. Li, FATDNet: A fusion adversarial network for tomato leaf disease segmentation under complex backgrounds, Comput. Electron. Agric. 234 (2025) 110270.

[41] A.N. Retta, D.H. Berhe, Tomato leaf miner–Tuta absoluta (Meyrick), a devastating pest of tomatoes in the highlands of Northern Ethiopia: A call for attention and action, Res. J. Agric. Environ. Manag. 4 (6) (2015) 264–269.

[42] N. Desneux, E. Wajnberg, K.A. Wyckhuys, G. Burgio, S. Arpaia, C.A. Narváez-Vasquez, J. González-Cabrera, D. Catalán Ruescas, E. Tabone, J. Frandon, et al., Biological invasion of European tomato crops by Tuta absoluta: ecology, geographic expansion and prospects for biological control, J. Pest Sci. 83 (2010) 197–215.

[43] D.P. Rubanga, L.K. Loyani, M. Richard, S. Shimada, A deep learning approach for determining effects of Tuta Absoluta in tomato plants, 2020, arXiv preprint arXiv:2004.04023.

[44] P.S. Georgantopoulos, D. Papadimitriou, C. Constantinopoulos, T. Manios, I. Daliakopoulos, D. Kosmopoulos, A multispectral dataset for the detection of Tuta absoluta and leveillula taurica in tomato plants, Smart Agric. Technol. 4 (2023) 100146.

[45] T. Uygun, M.M. Ozguven, Determination of tomato leafminer: Tuta absoluta (Meyrick)(Lepidoptera: Gelechiidae) damage on tomato using deep learning instance segmentation method, Eur. Food Res. Technol. 250 (6) (2024) 1837–1852.

[46] J. Cho, J. Choi, M. Qiao, C. Ji, H. Kim, K. Uhm, T. Chon, Automatic identification of whiteflies, aphids and thrips in greenhouse based on image analysis, Red 346 (246) (2007) 244.

[47] Y. Sun, H. Cheng, Q. Cheng, H. Zhou, M. Li, Y. Fan, G. Shan, L. Damerow, P.S. Lammers, S.B. Jones, A smart-vision algorithm for counting whiteflies and thrips on sticky traps using two-dimensional Fourier transform spectrum, Biosyst. Eng. 153 (2017) 82–88.

[48] Y. Qing, L. Jun, Q.-j. Liu, G.-q. Diao, B.-j. Yang, H.-m. Chen, T. Jian, An insect imaging system to automate rice light-trap pest identification, J. Integr. Agric. 11 (6) (2012) 978–985.

[49] W. Ding, G. Taylor, Automatic moth detection from trap images for pest management, Comput. Electron. Agric. 123 (2016) 17–28.

[50] G. Wei, S. Thomas, M. Cole, Z. Rácz, J.W. Gardner, Ratiometric decoding of pheromones for a biomimetic infochemical communication system, Sensors 17 (11) (2017) 2489.

[51] Z.-Q. Zhao, P. Zheng, S.-t. Xu, X. Wu, Object detection with deep learning: A review, IEEE Trans. Neural Netw. Learn. Syst. 30 (11) (2019) 3212–3232.

[52] J. Redmon, A. Farhadi, YOLO9000: better, faster, stronger, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 7263–7271.

[53] J. Redmon, A. Farhadi, Yolov3: An incremental improvement, 2018, arXiv preprint arXiv:1804.02767.

[54] A. Bochkovskiy, C.-Y. Wang, H.-Y.M. Liao, Yolov4: Optimal speed and accuracy of object detection, 2020, arXiv preprint arXiv:2004.10934.

[55] C. Li, L. Li, H. Jiang, K. Weng, Y. Geng, L. Li, Z. Ke, Q. Li, M. Cheng, W. Nie, et al., YOLOv6: A single-stage object detection framework for industrial applications, 2022, arXiv preprint arXiv:2209.02976.

[56] C.-Y. Wang, A. Bochkovskiy, H.-Y.M. Liao, YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 7464–7475.
[57] D. Reis, J. Kupec, J. Hong, A. Daoudi, Real-time flying object detection with YOLOv8, 2023, arXiv preprint arXiv:2305.09972.
[58] C.-Y. Wang, I.-H. Yeh, H.-Y. Mark Liao, Yolov9: Learning what you want to learn using programmable gradient information, in: European Conference on Computer Vision, Springer, 2024, pp. 1–21.
[59] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, G. Ding, Yolov10: Real-time end-to-end object detection, 2024, arXiv preprint arXiv:2405.14458.
[60] R. Khanam, M. Hussain, YOLOv11: An overview of the key architectural enhancements, 2024, arXiv preprint arXiv:2410.17725.
[61] R. Padilla, S.L. Netto, E.A. Da Silva, A survey on performance metrics for object-detection algorithms, in: 2020 International Conference on Systems, Signals and Image Processing, IWSSIP, IEEE, 2020, pp. 237–242.
[62] T.S. Mahdavi, H. Madadi, A. Biondi, Predation and reproduction of the generalist predator Nabis pseudoferus preying on Tuta absoluta, Èntomol. Exp. Appl. 168 (10) (2020) 732–741.
[63] M. Del Pino, J.R. Gallego, E. Hernández Suárez, T. Cabello, Effect of temperature on life history and parasitization behavior of Trichogramma achaeae Nagaraja and Nagarkatti (Hym.: Trichogrammatidae), Insects 11 (8) (2020) 482.
[64] K. Abbes, A. Biondi, A. Kurtulus, M. Ricupero, A. Russo, G. Siscaro, B. Chermiti, L. Zappala, Combined non-target effects of insecticide and high temperature on the parasitoid Bracon nigricans, PLoS One 10 (9) (2015) e0138411.
[65] A. Loni, K.G. Samartsev, P.L. Scaramozzino, S.A. Belokobylskij, A. Lucchi, Braconinae parasitoids (Hymenoptera, Braconidae) emerged from larvae of Lobesia botrana (Denis & Schiffermüller)(Lepidoptera, Tortricidae) feeding on Daphne gnidium L., ZooKeys (587) (2016) 125.
[66] L. Zappala, A. Biondi, A. Alma, I.J. Al-Jboory, J. Arno, A. Bayram, A. Chailleux, A. El-Arnaouty, D. Gerling, Y. Guenaoui, et al., Natural enemies of the South American moth, Tuta absoluta, in Europe, North Africa and Middle East, and their potential use in pest control strategies, J. Pest Sci. 86 (2013) 635–647.
[67] M.V. Balzan, F.L. Wäckers, Flowers to selectively enhance the fitness of a host-feeding parasitoid: adult feeding by Tuta absoluta and its parasitoid Necremnus artynes, Biol. Control 67 (1) (2013) 21–31.
[68] L. Dosdall, G. Gibson, O. Olfert, B. Keddie, B. Ulmer, Contributions to the life history, host range, and distribution of Necremnus tidius (Hymenoptera: Eulophidae), Ann. Entomol. Soc. Am. 100 (6) (2007) 861–868.
[69] M. Gebiola, U. Bernardo, A. Ribes, G.A. Gibson, An integrative study of Necremnus Thomson (Hymenoptera: Eulophidae) associated with invasive pests in Europe and North America: taxonomic and ecological implications, Zool. J. Linnean Soc. 173 (2) (2015) 352–423.
[70] S.M. Sayed, M.M. Montaser, G. Elsayed, S.A. Amer, Preliminary molecular identification of a predatory bug, Orius albidipennis, collected from ornamental plants, J. Insect Sci. 13 (1) (2013) 11.
[71] S. Bouagga, A. Urbaneja, J.L. Rambla, A. Granell, M. Pérez-Hedo, Orius laevigatus strengthens its role as a biological control agent by inducing plant defenses, J. Pest Sci. 91 (2018) 55–64.
[72] M.G. Tommasini, Collection of orius species in Italy, Bull. Insectology 57 (2) (2004) 65–72.
[73] J. Sanchez, A. Lacasa, J. Arnó, C. Castañé, O. Alomar, Life history parameters for Nesidiocoris tenuis (Reuter)(Het., Miridae) under different temperature regimes, J. Appl. Entomol. 133 (2) (2009) 125–132.
[74] E. Stepanycheva, M. Petrova, T. Chermenskaya, I. Shamshev, I. Pazyuk, The behavioral response of the predatory bug Orius laevigatus Fieber (Heteroptera, Anthocoridae) to synthetic volatiles, Èntomol. Rev. 94 (2014) 1053–1058.
[75] K.B. Ferguson, B.A. Pannebakker, A. Centurión, J. van Den Heuvel, R. Nieuwenhuis, F.F. Becker, E. Schijlen, A. Thiel, B.J. Zwaan, E.C. Verhulst, Bracon brevicornis genome showcases the potential of linked-read sequencing in identifying a putative complementary sex determiner gene, Genes 11 (12) (2020) 1390.